

CSSA HELL

Why would you do this

We have the following:

$$\begin{aligned}a_A &= -(g + kv) \\ a_B &= -(g + kv).\end{aligned}$$

Put $a = \frac{dv}{dt}$ and integrate.

$$\begin{aligned}\ln |g + kv_A| &= -kt + C_1 \\ \ln |g + kv_B| &= -kt + C_2.\end{aligned}$$

Set $t = 0$ for both, with initial unsigned velocities u_1 , u_2 , and take upwards as positive to yield:

$$\begin{aligned}C_1 &= \ln |g + ku_1| \\ C_2 &= \ln |g - ku_2|.\end{aligned}$$

Considering initial signs, we have:

$$\begin{aligned}v_A &= \left(\frac{g + ku_1}{k} \right) e^{-kt} - \frac{g}{k} \\ v_B &= - \left(\frac{ku_2 - g}{k} \right) e^{-kt} - \frac{g}{k}.\end{aligned}$$

Now, integrate for x_A , x_B :

$$\begin{aligned}x_A &= \int v_A dt \\ &= C_3 - \left(\frac{g + ku_1}{k^2} \right) e^{-kt} - \frac{gt}{k} \\ x_B &= \int v_B dt \\ &= C_4 + \left(\frac{ku_2 - g}{k^2} \right) e^{-kt} - \frac{gt}{k}.\end{aligned}$$

Taking initial conditions again, we have:

$$\begin{aligned}C_3 &= \frac{g + ku_1}{k^2} \\ C_4 &= H - \frac{ku_2 - g}{k^2},\end{aligned}$$

which gives

$$\begin{aligned}x_A &= (1 - e^{-kt}) \left(\frac{g + ku_1}{k^2} \right) - \frac{gt}{k} \\ x_B &= H - (1 - e^{-kt}) \left(\frac{ku_2 - g}{k^2} \right) - \frac{gt}{k}.\end{aligned}$$

Suppose $x_A(T)$ is maximised. Then, $x_B(T) - x_A(T) > 0$. Also,

$$T = \frac{1}{k} \ln \left(\frac{g + ku_1}{g} \right),$$

so

$$H - \left(\frac{ku_1}{g + ku_1} \right) \left(\frac{ku_2 - g}{k^2} \right) - \left(\frac{ku_1}{g + ku_1} \right) \left(\frac{g + ku_1}{k^2} \right) > 0.$$

Hence,

$$\begin{aligned} H &> \frac{u_1}{g + ku_1} \left(\frac{ku_2 - g}{k} \right) + \frac{u_1}{k} \\ &= \frac{ku_1u_2 + ku_1^2}{k(g + ku_1)} \\ &= \frac{ku_1(u_1 + u_2)}{k(g + ku_1)} \\ &= \frac{u_1 + u_2}{k} \left(1 - \frac{g}{g + ku_1} \right). \end{aligned}$$

However, for $t \neq 0$ and $x_A = 0$, we must have $x_B < 0$. Denote $t_0 \neq 0$ as the time when $x_A = 0$, so

$$\begin{aligned} \frac{gt_0}{k} &= (1 - e^{-kt_0}) \left(\frac{g + ku_1}{k^2} \right) \\ t_0 &= (1 - e^{-kt_0}) \left(\frac{g + ku_1}{gk} \right) \\ 1 - e^{-kt_0} &= \frac{g + ku_1}{gkt_0}. \end{aligned}$$

Substituting this into x_B , we get:

$$\begin{aligned} H &< (1 - e^{-kt_0}) \left(\frac{ku_2 - g}{k^2} \right) + (1 - e^{-kt_0}) \left(\frac{g + ku_1}{k^2} \right) \\ &< \left(\frac{ku_2 - g}{k^2} \right) + \left(\frac{g + ku_1}{k^2} \right) \\ &= \frac{u_1 + u_2}{k}. \end{aligned}$$

Combining these, we get

$$\boxed{\frac{u_1 + u_2}{k} \left(1 - \frac{g}{g + ku_1} \right) < H < \frac{u_1 + u_2}{k},}$$

and we are done.